

Vibe Shuffle: Browser-Based Affective Music Selection from Facial Valence and Cardiac Arousal

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June 19, 2026

Abstract

Vibe Shuffle is a client-only web application that selects music in response to a listener’s affective state, estimated in real time from facial expression and (optionally) heart-rate variability. All signal processing runs in the browser: no audio, video, or physiological data leaves the device, and only anonymous self-report ratings are exported. We describe the affect model—a fusion of a facial *v* channel and a cardiac *a* channel onto Russell’s valence–arousal circumplex [1]—and the track-ranking rule that maps the fused state to the next song. We then specify a blinded, counterbalanced validation protocol that contrasts Vibe Shuffle against a Random Shuffle baseline, with perceived *mood-fit* as the primary outcome and *liking* as a control. The system, this paper, and the data schema are openly available.

1 Introduction

Listeners routinely choose music to maintain or change how they feel [3]. Most streaming recommenders, however, optimise for long-run engagement from listening history rather than the listener’s *current* affective state. Vibe Shuffle asks a narrower question: if we estimate a listener’s momentary valence and arousal on-device and pick the next track to match that state, do listeners perceive the music as fitting their mood better than a random draw from the same pool?

The design priorities are (i) **privacy**—raw sensor streams never leave the browser; (ii) **falsifiability**—selection is evaluated against a matched random baseline under blinding; and (iii) **reproducibility**—the client, the curated track pool, and the export schema are public.

2 Affect Model

2.1 Signal chain

The pipeline is fully client-side:

camera → blendshapes → valence | heart rate → HR/RR → arousal → fusion → quadrant →
track ranking.

2.2 Valence from facial expression

A MediaPipe Face Landmarker [4] produces per-frame blendshape activations, which are reduced to *happy*, *tense*, and *sad* scores. A slowly learned personal neutral baseline is subtracted, and the positive-versus-negative balance is mapped to a continuous valence $v \in [0, 1]$. Head and

body motion (nose-tip drift plus frame differencing) feeds a motion channel that nudges the positive score upward and raises arousal.

2.3 Arousal from heart rate

When a Bluetooth heart-rate sensor is connected, HR and RR intervals are parsed into a 120s personal baseline. Short-window arousal is computed from the z -scored heart rate (higher HR $\rightarrow$ higher arousal) and the z -scored RMSSD (higher short-term variability $\rightarrow$ lower arousal) [5]:

$$a_{\text{phys}} = \text{clip}\left(\frac{1}{2} + \alpha z(\text{HR}) - \beta z(\text{RMSSD}), 0, 1\right). \quad (1)$$

SDNN is logged for quality control but excluded from the short-window estimate.

2.4 Fusion and quadrants

The fused state is

$$(v, a) = (v_{\text{face}}, a_{\text{phys}} + \gamma m), \quad (2)$$

where m is head motion and γ its weight. The face sets valence; a usable ECG sets the arousal base in both directions, and motion adds to arousal on top. Without a usable ECG, the face/motion channel carries arousal upward only; with neither face nor ECG, both axes default to 0.5. Splitting each axis at 0.5 yields four quadrants—Energetic, Calm, Tense, Melancholic—following the circumplex [1, 2].

2.5 Track selection

Each track in the fixed pool carries pre-computed valence/arousal coordinates (v_t, a_t) . In a Vibe block, the next track minimises distance to the fused target (v^*, a^*) , preferring the matching quadrant and penalising recently played tracks:

$$t^* = \arg \min_t \|(v_t, a_t) - (v^*, a^*)\|_2 + \lambda r(t), \quad (3)$$

where $r(t)$ is a recency penalty. A Random block draws from the same pool with a per-session seed, so order differs every run while the pool is held constant.

3 Validation Protocol

The app implements a blinded, counterbalanced within-subject comparison.

- **Pool:** 100 curated Spotify tracks, 25 per quadrant; identical for every participant.
- **Blocks:** two blocks, **random** and **vibe**, of 5 tracks each. Block order is randomised per session and recorded.
- **Listening window:** each track plays for 60s, then the rating prompt opens (early rating allowed).
- **Blinding:** the condition is never shown during the session.

After each track, two 5-point questions are asked in sequence: *liking* (“How much do you like this song?”) and *mood-fit* (“How well did it fit your current mood?”). Mood-fit is the primary outcome; liking is the control, separating “did not fit my mood” from “I just don’t like this song.”

Table 1: Selected exported fields (one row per track).

Field	Meaning
<code>block_mode</code>	<code>vibe</code> or <code>random</code>
<code>song_quadrant</code>	target affect quadrant of the track
<code>detected_valence/arousal</code>	fused listener state
<code>physiology_quality</code>	ECG signal-quality flag
<code>rating_fit_1_to_5</code>	primary outcome (mood-fit)
<code>rating_like_1_to_5</code>	control (liking)

4 Analysis Plan

The pre-specified comparison contrasts `rating_fit_1_to_5` between `vibe` and `random` blocks, controlling for `rating_like_1_to_5` and accounting for block order. Each completed session exports one CSV row per track with song features, the detected (v, a), signal-quality flags, and both ratings (see Table 1).

5 Privacy and Limitations

No backend exists: camera frames, blendshapes, and heart-rate packets are processed in memory and never transmitted; only the anonymous rating CSV leaves the browser, at the participant’s initiative. The estimator is a lightweight heuristic, not a clinical affect measure: facial valence is sensitive to lighting and occlusion, and the arousal channel requires a compatible BLE sensor. The track pool is small and fixed, so results speak to relative ranking within that pool rather than to catalogue-scale recommendation.

6 Availability

Source code: https://github.com/eybmits/vibe_shuffle. Live application: https://eybmits.github.io/vibe_shuffle_site/. This document is generated from `paper/vibe_shuffle.tex` in the source repository.

References

- [1] J. A. Russell. A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6):1161–1178, 1980.
- [2] J. Posner, J. A. Russell, and B. S. Peterson. The circumplex model of affect. *Development and Psychopathology*, 17(3):715–734, 2005.
- [3] P. N. Juslin and D. Västfjäll. Emotional responses to music: The need to consider underlying mechanisms. *Behavioral and Brain Sciences*, 31(5):559–575, 2008.
- [4] C. Lugaresi et al. MediaPipe: A framework for building perception pipelines. *arXiv:1906.08172*, 2019.
- [5] F. Shaffer and J. P. Ginsberg. An overview of heart rate variability metrics and norms. *Frontiers in Public Health*, 5:258, 2017.